THE DERIVED LIMIT OF AN INVERSE SYSTEM OF COMPLEXES FITS INTO THE CANONICAL PRODUCT DISTINGUISHED TRIANGLE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The computation of R lim and its first derived functor, including the short exact sequence from right acyclicity. invoked at proof lines 23284-23290 and 23294-23295. Supplies both the special-case sequence and the acyclic replacement in the reduction.
- Exactness of products in the category of abelian groups. used at proof lines 23296-23297. Transfers the quasi-isomorphism K -> L to the products.
- The derived-category definition of a derived limit of an inverse system. referred to at statement lines 23277-23279. Is the property that the displayed triangle is asserted to establish.

Source reference. Original lemma 08TC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11160–11160. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 22943: sources/stacks-more-algebra/pinned-original-source.tex, source lines 22943–22943. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let $K = (K_n^\bullet)$ be an object of $D(Ab(\mathbf{N}))$. There exists a canonical distinguished triangle*

$$R\lim K \to \prod\nolimits_n K_n^\bullet \to \prod\nolimits_n K_n^\bullet \to R\lim K[1]$$

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in $D(Ab)$. In other words, $R\lim K$ is a derived limit of the inverse system $(K_n^\bullet)$ of $D(Ab)$, see Derived Categories, Definition 08TC.

Proof. Suppose that for each p the inverse system (K_n^p) is right acyclic for lim. By Lemma 22943 this gives a short exact sequence

$$0 \to \lim_n K_n^p \to \prod\nolimits_n K_n^p \to \prod\nolimits_n K_n^p \to 0$$

for each p. Since the complex consisting of $\lim_n K_n^p$ computes $R\lim K$ by Lemma 22943 we see that the lemma holds in this case.

Next, assume $K = (K_n^\bullet)$ is general. By Lemma 22943 there is a quasi-isomorphism $K \to L$ in $D(Ab(\mathbf{N}))$ such that (L_n^p) is acyclic for each p. Then $\prod K_n^\bullet$ is quasi-isomorphic to $\prod L_n^\bullet$ as products are exact in Ab, whence the result for L (proved above) implies the result for K. □